# Series: Method of Differences – Part 2

Questions 1–18

**Question 1.** Show that

$$\sum_{r=1}^{n} \frac{1}{r(r+1)} = \frac{n}{n+1}.$$

**Question 2.** Use the method of differences to find the value of $\sum_{r=1}^{40} \frac{1}{r(r+1)(r+2)}$, giving your answer in exact form.

**Question 3.** Use the method of differences to find the value of $\sum_{r=10}^{200} \frac{1}{r^2+5r+6}$, giving your answer in exact form.

**Question 4.** Use the method of differences to determine the value of $\sum_{r=5}^{55} \frac{1}{(r+1)(r+2)(r+3)}$, giving your answer in exact form.

**Question 5.** Show that

$$\sum_{r=10}^{n} \frac{1}{(r+3)(r+4)} = \frac{n-a}{b(n+c)},$$

where $a$, $b$ and $c$ are integers to be found.

**Question 6.** Use the method of differences to prove that for $n > 2$

$$\sum_{r=2}^{n} \frac{4}{4r^2-4r-3} = \frac{a(n-b)(an+b)}{cn^2-d},$$

where $a$, $b$, $c$ and $d$ are integers to be found.

**Question 7.**

a) Find an expression for $\sum_{r=1}^{n} \frac{2}{r^2+4r+3}$.

b) Hence determine the value of $\sum_{r=1}^{\infty} \frac{1}{r^2+4r+3}$.

**Question 8.** Find $\sum_{r=1}^{\infty} \frac{1}{(2r+1)(2r+3)}$.

**Question 9.**

a) Find an expression for $\sum_{r=1}^{n} \frac{3}{r^2+3r}$.

b) Hence determine the value of $\sum_{r=1}^{\infty} \frac{1}{r^2+3r}$.

**Question 10.** Find the smallest integer $n$ such that $\sum_{r=3}^{n} \frac{1}{r^2-3r+2} > 0.45$.

**Question 11.**

a) Show that $\sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} = \frac{an+b}{(3n-2)(6n+1)}$, where $a$ and $b$ are integers to be found.

b) Hence find $\lim_{n\to\infty} \left( \sum_{r=n}^{2n} \frac{1}{(3r-2)(3r+1)} \right)$.

**Question 12.** A company is managing a reservoir with a total capacity of 5 million litres of water. To regulate water usage, the company models the amount of water released each year as follows:

$$W = \frac{15}{r^2+6r+8},$$

where $W$ is the amount of water, in millions of litres, released in year $r$, where $r = 1$ represents year one.

a) Show that the total amount of water released in the first $n$ years is given by $\frac{an^2+bn}{c(n+3)(n+4)}$, where $a$, $b$ and $c$ are constants to be found.

b) Explain why, according to this model, the reservoir will never run out of water.

c) The company decides to release an additional fixed amount of water each year to ensure that the entire 5 million litres is used up exactly after 30 years. Refine the formula for $W$ to reflect this change.

**Question 13.** A government agency releases funding to startup businesses over time. The amount released in month $r$ measured in millions of pounds (£) is given by

$$A_r = \frac{300}{(r+1)(r+4)}, \quad r = 1, 2, 3, \ldots,$$

where $A_r$ is the amount of funding released in month $r$.

a) Show that the total amount of funding released in the first $n$ months can be written in the form $A = \dfrac{25n(an^2 + bn + c)}{3(n+2)(n+3)(n+4)}$, where $a$, $b$ and $c$ are constants to be found.

b) Explain why, according to this model, the government agency expects not to release more than £109 million in total.

c) The agency claims that after 30 months, at least 90% of the total funding will have been released. Verify this claim.

**Question 14.** Let $C_r = \dfrac{\sin 1}{\cos r \cos(r+1)}$, where $r$ is a positive integer.

a) By considering $\sin r \cos(r+1) - \cos r \sin(r+1)$, show that $C_r = \tan(r+1) - \tan r$.

b) Hence evaluate $\displaystyle\sum_{k=1}^{n} C_r$.

**Question 15.** Let $D_r = \dfrac{\sinh 1}{\sinh(r)\sinh(r+1)}$, where $r$ is a positive integer.

a) By considering $\sinh(r+1)\cosh r - \cosh(r+1)\sinh r$, show that $D_r = \coth r - \coth(r+1)$.

b) Hence evaluate $\displaystyle\sum_{k=1}^{n} D_r$.

**Question 16.**

a) Use $\tan(A-B) = \dfrac{\tan A - \tan B}{1+\tan A \tan B}$ with $A = (r+1)x$ and $B = rx$ to show that

$$\tan(rx)\tan((r+1)x) = \frac{\tan((r+1)x)}{\tan x} - \frac{\tan(rx)}{\tan x} - 1.$$

b) Hence show that

$$\tan\frac{2\pi}{20}\tan\frac{3\pi}{20} + \tan\frac{3\pi}{20}\tan\frac{4\pi}{20} + \cdots + \tan\frac{14\pi}{20}\tan\frac{15\pi}{20} = \frac{\tan\frac{15\pi}{20}}{\tan\frac{\pi}{20}} - 15.$$

**Question 17.** A sequence $u_r$ is defined by $u_r = \ln(r+1)$, where $r \geq 1$.

a) Show that $u_{r+1} - u_r = \ln(1 + \frac{1}{r+1})$.

b) Show that $u_{r+1} - u_r = \displaystyle\int_{r+1}^{r+2} \frac{1}{x}\, dx$ and hence, using the Mean Value Theorem for integrals, show that $\dfrac{1}{r+2} < u_{r+1} - u_r < \dfrac{1}{r+1}$.

c) Using method of differences, show that $\displaystyle\sum_{r=1}^{n}(u_{r+1} - u_r) = \ln(n+2) - \ln 2$.

**Question 18.** A sequence is defined by $u_r = y(r)$ where $y(x)$ satisfies the differential equation $\dfrac{dy}{dx} = \dfrac{1}{x(x+1)}$ for $x > 0$.

a) Find the general solution to the differential equation.

b) Hence find an expression for $u_{r+1} - u_r$.

c) Use the method of differences to evaluate $\displaystyle\sum_{r=1}^{n}(u_{r+1} - u_r)$, giving your answer as a single term.

d) Hence, find $\displaystyle\sum_{r=1}^{\infty}(u_{r+1} - u_r)$.

# Maclaurin Series - Part 2

Questions 1–46

**Question 1.** Find $f'(x)$, $f''(x)$, $f^{(3)}(x)$ and $f^{(4)}(x)$ for each of the following:

a) $f(x) = \sin x$
b) $f(x) = \cos x$
c) $f(x) = e^x$
d) $f(x) = \ln(1 + x)$
e) $f(x) = (1 + x)^n$ for constant $n$

**Question 2.** For each of the following, derive the Maclaurin series by using up to and including $f^{(4)}(x)$:

a) $f(x) = \sin x$
b) $f(x) = \cos x$
c) $f(x) = e^x$
d) $f(x) = \ln(1 + x)$
e) $f(x) = (1 + x)^n$ for constant $n$

**Question 3.** By considering differentials, explain why the Maclaurin series for $\ln x$ does not exist.

**Question 4.** For each of the following, use differentiation to find the first three non-zero terms in the Maclaurin series and state the range of values for which the expansion is valid:

a) $e^{5x}$
b) $\sin 3x$
c) $\cos \frac{x}{2}$
d) $\ln(1+4x)$
e) $\ln(x+2)$
f) $(1+2x)^{1/4}$
g) $\dfrac{1}{(1-2x)^3}$

**Question 5.** Derive the first three non-zero terms of the Maclaurin series for:

a) $\cosh x$

b) $\sinh x$

**Question 6.** It is given that $f(x) = \sqrt{1+4x}$.

a) Find $f'(x)$ and $f''(x)$.

b) Determine the first three terms of the Maclaurin series for $\sqrt{1+4x}$.

c) State the validity of the Maclaurin series for $\sqrt{1+4x}$.

d) Hence, using a suitable value for $x$, show that $\sqrt{6} \approx \frac{39}{16}$.

**Question 7.**

a) Use differentiation to find the first three non-zero terms in the Maclaurin series for $\ln\left(\dfrac{x+2}{x+1}\right)$.

b) Determine the values for which the expansion is valid.

**Question 8.** It is given that

$$\sin\left(x-\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}\left(-1+x+\frac{1}{2}x^2-\frac{1}{6}x^3-\frac{1}{24}x^4+\cdots\right).$$

a) Use the series expansions of $\sin x$ and $\cos x$ to prove the result.

b) Use differentiation to prove the result.

**Question 9.** It is given that

$$\cos\left(x+\frac{3\pi}{4}\right)=\frac{\sqrt{2}}{2}\left(-1-x+\frac{1}{2}x^2+\frac{1}{6}x^3-\frac{1}{24}x^4+\cdots\right).$$

a) Use the series expansions of $\sin x$ and $\cos x$ to prove the result.
b) Use differentiation to prove the result.

**Question 10.**

a) Using differentiation, find the Maclaurin series of $\sin 2x$ up to and including the $x^3$ term.
b) Using differentiation, find the Maclaurin series of $\cos 2x$ up to and including the $x^4$ term.
c) Hence, given $x$ is small such that terms in $x^5$ and higher powers are negligible, show that

$$\cos\left(2x+\frac{\pi}{6}\right)\approx\frac{\sqrt{3}}{2}-x-\sqrt{3}x^2+\frac{2x^3}{3}+\frac{\sqrt{3}x^4}{3}.$$

**Question 11.**

a) Use differentiation to find the Maclaurin series for $\sin\left(\frac{\pi}{3}+x\right)$ up to and including the $x^2$ term.

b) Hence show that when $h$ is small,

$$\int_0^h x^2 \sin\left(\frac{\pi}{3}+x\right) dx \approx \frac{\sqrt{3}h^3}{6} + \frac{h^4}{8} - \frac{\sqrt{3}h^5}{20}.$$

**Question 12.** Use the Maclaurin series for $\cos 2x$ to find the series expansion for $\cos^2 x$ up to and including the $x^6$ term.

**Question 13.**

a) Use the Maclaurin series expansion for $\cos x$ to find the series expansion of $\sin^2(\frac{x}{2})$ up to and including the $x^4$ term.

b) Hence find an approximation to $\displaystyle\int_{\pi/3}^{\pi/2} \left(\frac{1}{x}\sin^2(\frac{x}{2})\right) dx$, giving your answer to two decimal places.

**Question 14.**

a) Use the Maclaurin series for $\cos 2x$ to find the series expansion for $\sin^2 x$ up to and including the $x^6$ term.

b) Hence, using your result from part (a), find an approximation for $\displaystyle\int_0^{\pi/2} \sin^2 x\, dx$.

c) Evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x\, dx$.

d) Find the percentage error in your approximation, giving your answer to two significant figures.

**Question 15.** It is given that $f(x) = \ln(1 - x + \sin x)$ and that the constant $k$ lies in the domain of $f$.

a) Show that $k$ can never be greater than 2.

b) Find $f'(x)$.

c) By using $\cos x \approx 1 - \frac{1}{2}x^2$ and $\sin x \approx x$, determine a cubic approximation for $f(x)$ for small values of $x$.

**Question 16.** It is given that $f(x) = \cosh^2 x$.

a) Show that $f''(x) = 2\cosh(2x)$.

b) Explain why the coefficients of the odd powers of $x$ in the Maclaurin series for $\cosh^2 x$ are all zero.

c) Given $n$ is a positive even number, find the coefficient of $x^n$ in this series.

**Question 17.** It is given that $f(x) = \sinh^n x$ for $n \geq 2$.

a) By using differentiation find the first three non-zero terms of the Maclaurin series of $\sinh(x)$.

b) Hence, find the first three non-zero terms of the Maclaurin series of $\sinh^n(x)$.

**Question 18.**

a) Use differentiation to find the first two non-zero terms of the Maclaurin expansion of $\ln(1 + \cos x)$.

b) By considering the root of the equation $\ln(1 + \cos x) = 0$, deduce that $\pi \approx 2\sqrt{2}$.

**Question 19.** Use differentiation to find the first two non-zero terms of the Maclaurin series of $\sec x$.

**Question 20.**

a) Use differentiation to show that the first three non-zero terms of the Maclaurin series of $\cos x$ are $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$.

b) Hence, given that $x$ is small, find $a$ and $b$ if $\sec x = 1 + ax^2 + bx^4$.

**Question 21.**

a) Use differentiation to show that the first three non-zero terms of the Maclaurin series of $\sin x$ are $x - \frac{1}{6}x^3 + \frac{1}{120}x^5$.

b) Hence, given that $x$ is small, find $a$ and $b$ if $\operatorname{cosec} x = \frac{1}{x} + ax + bx^3$.

**Question 22.** It is known that $f(x) = \sin(x^3)$.

a) Find $f'(x)$.

b) Use a series expansion to find an approximation to $f'(x)$ in terms up to and including the $x^{20}$ term.

c) Integrate your answer in part (b).

d) State the series expansion of $\sin(x^3)$ as far as the term in $x^{21}$.

**Question 23.** It is given that $f(x) = (1-x)^2 e^x$.

a) Show that $f''(x) = (x^2 + 2x - 1)e^x$.

b) Hence write $(1-x)^2 e^x$ in ascending powers of $x$ up to and including the $x^2$ term.

**Question 24.** It is given that $f(x) = (1-x)\sin x$.

a) Show that $f''(x) = (x-1)\sin x - 2\cos x$.

b) Hence write $(1-x)\sin x$ in ascending powers of $x$ up to and including the term in $x^3$.

**Question 25.** It is given that $f(x) = e^{2x} \sin x$.

a) Find $f(0)$ and $f'(0)$.

b) Show that $f''(x) = 4f'(x) - 5f(x)$.

c) Hence find the Maclaurin series for $f(x)$ up to and including the $x^3$ term.

**Question 26.** It is given that $f(x) = e^{-x} \cos x$.

a) Find $f(0)$ and $f'(0)$.

b) Show that $f''(x) = -2f'(x) - 2f(x)$.

c) Hence find the Maclaurin series for $f(x)$ up to and including the $x^3$ term.

**Question 27.** It is given that $f(x) = \sin x \cosh x$ and $f^{(4)}(x) = kf(x)$, where $k$ is an integer.

a) Find the value of $k$.

b) Hence determine the first three non-zero terms of the Maclaurin series for $f(x)$.

**Question 28.** Determine the Maclaurin series of $e^{x^2+1}$ up to and including the $x^3$ term:

a) Using differentiation.

b) Without using differentiation.

**Question 29.** It is given that $f(x) = \dfrac{1}{1+e^{-x}}$.

a) Show that $f''(x) = \dfrac{e^x(1-e^x)}{(e^x+1)^3}$.

b) Hence find the first three non-zero terms of the Maclaurin series of $\dfrac{1}{1+e^{-x}}$.

**Question 30.**

a) Show that $\dfrac{d^3}{dx^3}(\tan x) = 2\sec^2 x(\sec^2 x + 2\tan^2 x)$.

b) Use differentiation to find the Maclaurin series for $\tan x$ up to and including the term in $x^3$.

c) Hence show that when $h$ is small, $\displaystyle\int_h^{3h} \frac{\tan x}{2x}\,dx \approx h + \frac{13}{9}h^3$.

**Question 31.** It is given that $f(x) = \ln(\cos x)$.

a) Show that $f'(x) = -\tan x$.

b) Show that $\dfrac{d}{dx}(\sec^2 x) = 2\tan x \sec^2 x$.

c) Write $\ln(\cos x)$ as a Maclaurin series in ascending powers of $x$ up to and including the $x^4$ term.

d) Hence find an approximation for $\ln(\frac{\sqrt{3}}{2})$.

e) Find the percentage error in the approximation, giving your answer to three significant figures.

**Question 32.**

a) Show that $\dfrac{d}{dx}(\ln(\sin x)) = \cot x$.

b) Hence, by using the first three terms of the Maclaurin series for $\sin x$ and $\ln(1+x)$, find the first three non-zero terms in the expansion of $\cot x$.

**Question 33.**

a) Show that $\frac{d}{dx}(\ln(\tan(\frac{x}{2}))) = \operatorname{cosec} x$.

b) Given that $\tan x \approx x + \frac{x^3}{3} + \frac{2x^5}{15}$, find the first three non-zero terms in the expansion of $\operatorname{cosec} x$.

**Question 34.** It is given that $f(x) = \arcsin x$.

a) Show that $f'(x) = \frac{1}{\sqrt{1-x^2}}$.

b) Hence find the first two non-zero terms of the Maclaurin series for $\arcsin x$.

**Question 35.** It is given that $f(x) = \arccos x$.

a) Show that $f'(x) = -\dfrac{1}{\sqrt{1-x^2}}$.

b) Hence find the first three terms of the Maclaurin series for $\arccos x$.

**Question 36.** It is given that $f(x) = \arctan x$.

a) Show that $f'(x) = \dfrac{1}{1+x^2}$.

b) Hence find the first two non-zero terms in the Maclaurin expansion of $\arctan x$.

c) Show that $\dfrac{\pi}{4} = 1 - \dfrac{1}{3} + \cdots$.

**Question 37.**

a) Find the first four non-zero terms of the Maclaurin series for $y = \dfrac{1}{\sqrt{1+x^2}}$.

b) Hence find the first four non-zero terms of the Maclaurin series for $y = \operatorname{arsinh} x$.

**Question 38.** Show that the first four non-zero terms of the Maclaurin series for $\operatorname{artanh} x$ are given by $x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \dfrac{x^7}{7}$.

**Question 39.**

a) Find the first three terms of the Maclaurin series for $\arccos x$.

b) Show that the Maclaurin series expansion for $(\arccos x)\ln(1+x)$ up to and including the $x^3$ term is $\frac{\pi}{2}x+(-1-\frac{\pi}{4})x^2+\frac{1}{6}(3+\pi)x^3$.

**Question 40.**

a) Use differentiation to find the Maclaurin series for $\arctan(1+x)$ up to and including the $x^2$ term.

b) Hence show that when $h$ is small, $\int_{-h}^{h} x\arctan(1+x)\,dx \approx \frac{1}{3}h^3$.

**Question 41.**

a) Find the first two terms of the Maclaurin series for $\arcsin x$.

b) Hence find an approximation to $\int_0^{1/2} \arcsin x \, dx$, giving your answer to three significant figures.

c) Determine an exact value of $\int_0^{1/2} \arcsin x \, dx$.

d) Find the percentage error in your approximation, giving your answer to one decimal place.

**Question 42.** It is given that $f(x) = \arcsin(x + \frac{1}{3})$.

a) Find constants $a$ and $b$ if the Maclaurin expansion of $f(x)$ is $\arcsin(\frac{1}{3}) + ax + bx^2 + \cdots$.

b) Hence find an approximate value of $\int_0^{0.05} f(x) \, dx$, giving your answer to three significant figures.

**Question 43.** It is given that $f(x) = \arccos(2x)$.

a) Find $f'(x)$.

b) By using a standard series to expand $f'(x)$, find the series for $f(x)$ up to and including the term in $x^5$.

**Question 44.**

a) Derive the Maclaurin series of $\arcsin(2x)$ up to and including the $x^3$ term.

b) By substituting $x = \frac{1}{4}$, show that $\pi \approx 6(\frac{1}{2} + \frac{1}{48})$.

c) Find the percentage error in your answer to part (b).

**Question 45.** It is given that $f(x) = \text{artanh}(\frac{1-x}{3+x})$ for $x > -1$.

a) Show that $f'(x) = -\dfrac{1}{2x+2}$.

b) Find $f''(x)$ in its simplest form.

c) Hence show that the first three non-zero terms of the Maclaurin series for $f(x)$ are $\dfrac{1}{2}\ln 2 - \dfrac{1}{2}x + \dfrac{1}{4}x^2$.

**Question 46.** It is given that $f(x) = \arctan(\cosh x)$.

a) Show that $f''(x) = \dfrac{\cosh x(1 + \cosh^2 x - 2\sinh^2 x)}{(1+\cosh^2 x)^2}$.

b) Find the Maclaurin series for $f(x)$ up to and including the $x^2$ term.